

AMAN BANSAL

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PROFESSIONAL SUMMARY

Application Support and AI-Native Software Engineer with 2+ years of experience shipping production-grade applications using Python, React, and LLM APIs. Dual Post-Graduate Certificates in Artificial Intelligence and Cybersecurity from Durham College. Experienced in building agentic pipelines, RAG systems, and full-stack production systems end-to-end, with a strong focus on code quality, testability, incident management, and pragmatic judgment about when AI adds genuine value. Interest in financial technology and capital markets systems. Eligible to work in Canada.

SKILLS

Programming Languages: Python, JavaScript, TypeScript, Node.js, React, Next.js, Bash/Shell scripting, Java

AI / LLM and Agentic AI: Claude API, OpenAI API, prompt engineering, prompt optimization, context management, LLM tool use, agentic pipeline design, Retrieval-Augmented Generation (RAG), vector stores, semantic search, embedding pipelines, agent evaluation

AI Development Tools: Cursor, Claude Code, GitHub Copilot: AI-native development workflows for accelerated prototyping and production delivery

Frontend: React, Next.js, Tailwind CSS, shadcn/ui, JavaScript, TypeScript, responsive UI design

Backend and Databases: Node.js, FastAPI, Python, REST API design, SQL, PostgreSQL, Supabase, MongoDB, Oracle, MSSQL, authentication, serverless backend patterns

Cloud and Deployment: AWS (EC2, VPC, S3, Security Groups), GCP (Cloud Run, Artifact Registry), Vercel, Docker, serverless deployment, Linux (Ubuntu/WSL)

DevOps and CI/CD: GitHub Actions, Docker, multi-OS parallel testing, automated release pipelines, post-release validation, MLOps practices

Monitoring and Observability: Splunk, Prometheus, Grafana, AppDynamics (familiar), SIEM event analysis

Support Tooling: ServiceNow (familiar), ITIL incident and change management principles, problem management workflows

Security: Vulnerability assessment, risk and vulnerability identification, network security architecture (VPCs, firewalls, NACLs), OWASP principles, penetration testing, secure API design

EXPERIENCE

Marketing Assistant and Campus Tour Guide

May 2025 – May 2026

Durham College, Oshawa, ON

- Collaborated cross-functionally with multiple departments to coordinate large-scale recruitment events including Open House and Program Information Nights; provided in-office IT support covering OS troubleshooting, networking, and classified student data management across internal drive and documentation systems.
- Maintained a structured student recruitment database tracking applications, registrations, and high school board contacts across Canada; coordinated communication workflows between the recruitment team and partner high schools to streamline campus visit scheduling and event reservations.
- Produced and edited promotional video content across all major social media platforms; monitored engagement analytics on Durham College's official pages to track performance and inform content decisions.

Website Manager

Dec 2022 – July 2023

Apexsupplements.in

- Provided day-to-day production support for a live e-commerce web application, triaging issues, deploying fixes with zero downtime, and implementing performance improvements that directly improved user experience and business outcomes.
- Wrote and optimized SQL queries to analyze product and customer data, producing actionable insights and documenting technical procedures and support runbooks for the broader team.

Full-Stack Developer

Jun 2021 – Nov 2021

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- Built and deployed a full-stack e-commerce platform (React, Node.js, MongoDB, JWT) end-to-end, from architecture and backend API design through frontend implementation, production release, and post-release checkout validation.
- Provided ongoing production support, identifying and resolving application vulnerabilities including authentication weaknesses and input validation gaps; documented remediation procedures applying security best practices to a client-facing system.

PROJECTS

Agentic RAG Pipeline, Institutional Knowledge Assistant *Python, React, GCP Cloud Run, MongoDB, GitHub Actions, Claude API, OpenAI-compatible LLMs, LangChain, LangGraph*

- Designed and shipped a full-stack LLM-powered agentic RAG application end-to-end using Claude API and Cursor; engineered prompt templates, iterative prompt optimization, context management strategies, and tool-use workflows to improve retrieval accuracy, with a complete evaluation framework measuring accuracy and relevance metrics.
- Implemented semantic search with vector store embeddings, semester-aware query logic, document storage on GCP, full CI/CD via GitHub Actions with 25 unit tests, and an ETL ingestion pipeline; made deliberate tradeoffs throughout between LLM-driven and traditional logic based on latency, cost, and reliability requirements.

Social Media Integrity Platform *Python, DistilBERT, Random Forest, FastAPI, Docker, React*

- Led AI/ML development on a 6-person cross-functional team delivering a scalable misinformation and bot detection platform; trained transformer NLP and ensemble ML models on 700,000+ samples, deployed via containerized FastAPI with rigorous evaluation metrics, demonstrating large-scale AI application delivery from prototype to production.

Fraud Detection API *Python, XGBoost, FastAPI, Scikit-learn, Docker, PostgreSQL*

- Built and deployed a fraud detection API achieving 95.62% F1-score on imbalanced financial data; designed backend API with explainability output for financial services decision-making, production-grade, scalable, and auditable by design.

Cybersecurity and SIEM Projects *Nessus, Snort, Splunk, Metasploit, Burp Suite, Isolation Forest, Nmap*

- Designed a secure virtual network environment with vulnerability scanning, IDS/IPS rules, and SIEM event capture via Splunk; conducted penetration testing engagements covering SQL injection, XSS, and authentication flaws with documented remediation reports.
- Built an ML-powered SIEM anomaly detector (Isolation Forest) on network flow data for risk and vulnerability identification; performed disaster recovery testing scenarios and validated alerting pipelines to ensure system resilience.

EDUCATION / CERTIFICATION

Durham College, Oshawa

Post Grad Cert, Artificial Intelligence Analysis, Design and Implementation

Sept 2025 – April 2026

Durham College, Oshawa

Post Grad Cert, Cyber Security

Sept 2024 – April 2025

GL Bajaj Institute of Management and Technology, Greater Noida

Bachelors in Computer Application (BCA)

Jul 2020 – Jul 2023

Front-End Development, Coursera

HTML/CSS/JavaScript, TypeScript, Bootstrap, React, Tailwind CSS, GitHub version control

June 2023 – Aug 2023

Foundation of Cyber Security, Coursera

Cybersecurity practices, threat identification, SIEM tools

Sep 2023 – Nov 2023

Network Security, Coursera

Network defense, intrusion tactics, system hardening, packet analysis

Dec 2023 – Feb 2024